

ASSIGNMENT - (2)

Q1). Convert the following :-

(a). $(6237.4501)_8 \rightarrow (\dots)_{10}$

Step ① :-

6	2	3	7	4	5	0	1
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Step ② :-

8^3	8^2	8^1	8^0	8^{-1}	8^{-2}	8^{-3}	8^{-4}
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Step ③ :- $6 \times 8^3 + 2 \times 8^2 + 3 \times 8^1 + 7 \times 8^0 + 4 \times \frac{1}{8} + 5 \times \frac{1}{8^2} +$

$0 \times \frac{1}{8^3} + 1 \times \frac{1}{8^4}$

$(10110010001)_2 = 3072 + 128 + 24 + 7 + \frac{1}{2} + \frac{5}{64} + 0 + \frac{1}{4096}$

$= 3231 + 0.5 + 0.0781 + 0.0002$

$= 3231.5783$

$(6237.4501)_8 = (3231.5783)_{10}$

(b). $(5328.625)_{10} \rightarrow (\dots)_8$

Integer Part $\Rightarrow$

8	5328	
8	666	0
8	83	2
8	10	3
	1	2

$= 12320$

Fractional Part $\Rightarrow 0.625 \times 8 = 5.000 \rightarrow 5$

First octal fractional digit = 5 ; remainder = 0 $\rightarrow$ stops

$(5328.625)_{10} \rightarrow (12320.5)_8$

ASSIGNMENT-2

(c). $(1101)_{10} = (\dots)_2$

2	1101	1
2	550	0
2	275	0
2	137	1
2	68	1
2	34	0
2	17	0
2	8	1
2	4	0
2	2	0
2	1	0
2	0	1

$(1101)_{10} = (10001001101)_2$

(d). $(B91D)_{16} = (\dots)_{10}$

$$= 11 \times 16^3 + 9 \times 16^2 + 1 \times 16^1 + 13 \times 16^0$$

$$= 11 \times 4096 + 9 \times 256 + 1 \times 16 + 13 \times 1$$

$$= 45056 + 2304 + 16 + 13$$

$$= 47389$$

$(B91D)_{16} = (47389)_{10}$

(e). $(3F)_{16} = (\dots)_2$

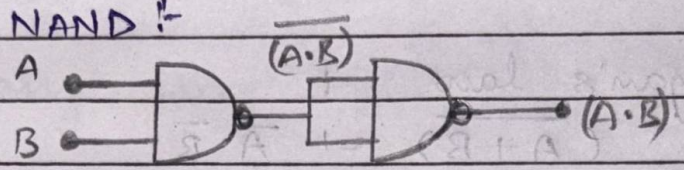
$\Rightarrow (3)_{16} = (0011)_2 \quad ; \quad \Rightarrow (F)_{16} = (1111)_2$

$(3F)_{16} = (00111111)_2$

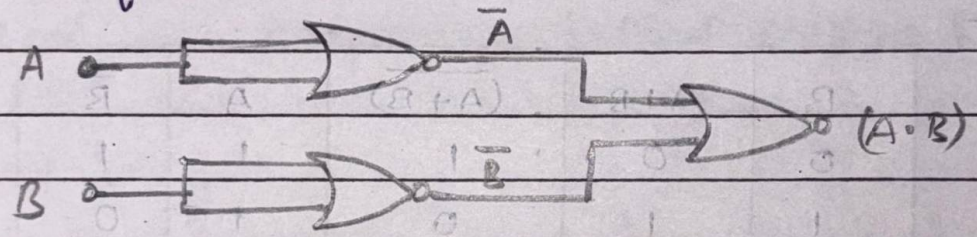
Q2). What are universal logic gates? Implement AND and OR gates using universal gates?

Ans2). NAND and NOR are universal logic gates because one can create each and every possible logic gate with them.

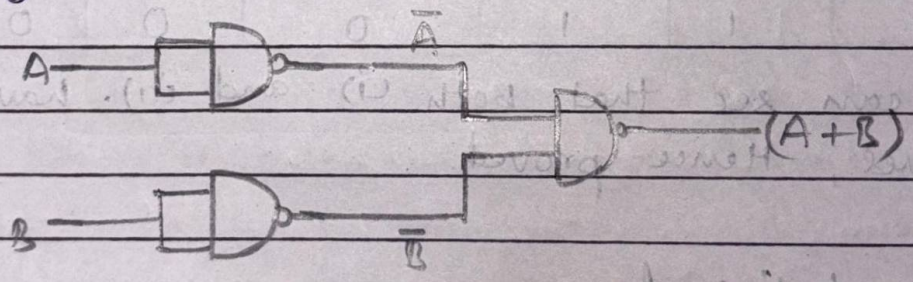
⇒ AND from NAND :-



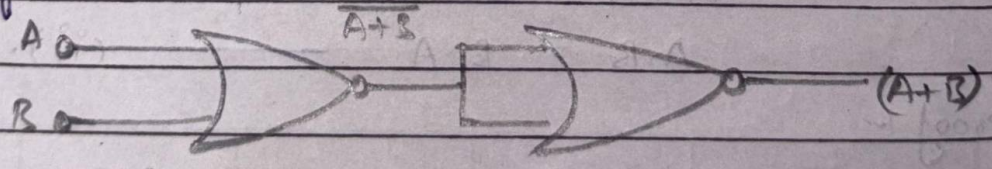
⇒ AND from NOR :-



⇒ OR from NAND :-



⇒ OR from NOR :-



Q3) What do you understand by Boolean algebra?
State and prove.

(Q) De-Morgan's Theorem.

Ans: Boolean algebra is a mathematical system that deals with binary variable having values 0 and 1 and logical operation AND, OR, NOT, etc.

⇒ De-Morgan's law

$$\overline{(A+B)} = \bar{A} \cdot \bar{B}$$

$$\overline{A \cdot B} = \bar{A} + \bar{B}$$

Proof 1-

A	B	A+B	$\overline{(A+B)}$	$\bar{A}$	$\bar{B}$	$\bar{A} \cdot \bar{B}$
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0

One can see that both (i) and (ii) have same values, Hence proved.

(Q) Commutative law.

Ans:

$$A+B = B+A \quad \text{--- (i)}$$

$$A \cdot B = B \cdot A \quad \text{--- (ii)}$$

Proof 1-

A	B	A+B	B+A	A·B	B·A
0	0	0	0	0	0
0	1	1	1	0	0
1	0	1	1	0	0
1	1	1	1	1	1

One can see both (i) and (ii) have same set of values in truth table - Hence Proved.

Q4): Explain half adder? Implement the full adder using two half adder.

Ans4): Half adder is a basic digital circuit that add the 2 single bit binary input (A and B) to produce a sum (S) and a carry (C), where the sum is XOR of the input, it doesn't consider any and the carry is the AND of the input, it doesn't consider any carry input from previous addition.

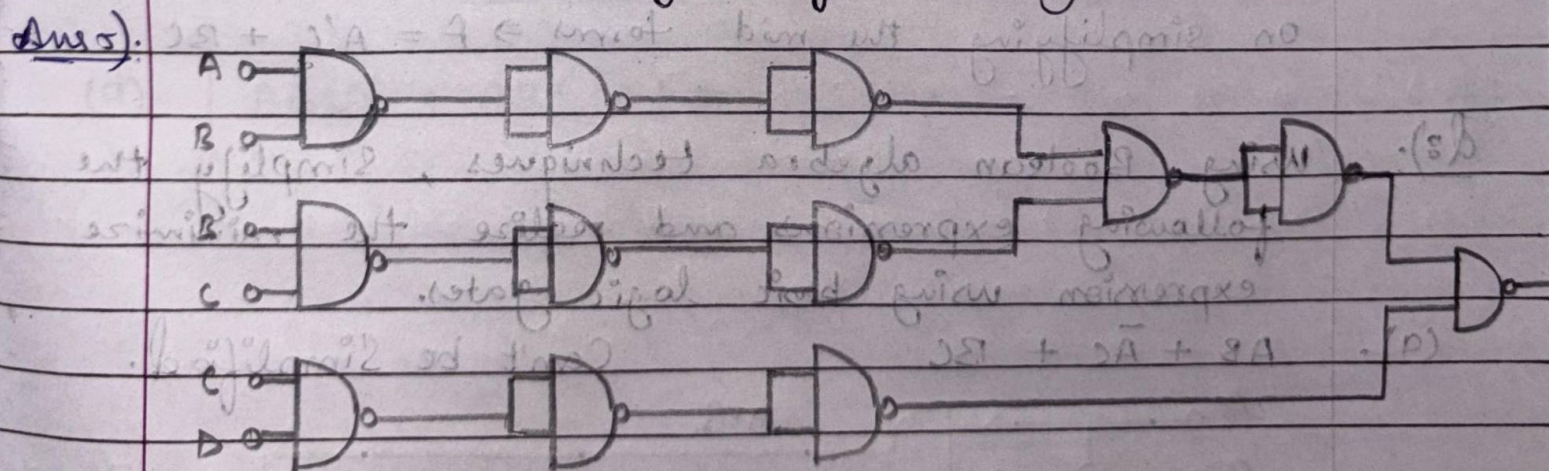
1). First half adder $\left\{ \begin{array}{l} \text{Sum 1} = A \oplus B \\ \text{Carry 1} = A \cdot B \end{array} \right\}$

2). Second half adder $\left\{ \begin{array}{l} \text{Final Sum} = \text{Sum 1} \oplus \text{Cin} \\ \text{Carry 2} = \text{Sum 1} \cdot \text{Cin} \end{array} \right\}$

3). Final carry $\text{Carry} = \text{Carry 1} + \text{Carry 2}$

Thus, the full adder sum could be made from half adder.

Q5): Draw the diagram of digital circuit for $F(A, B, C, D) = AB + BC + CD$ using only NAND gate.



Q6). Minimize the Boolean function using K-map.

$$F(A, B, C) = \sum m(0, 1, 2, 4, 6)$$

Ans 6).

A \ BC	00	01	11	10
0	1	1	0	1
1	1	0	0	1

$$M_0 = 000$$

$$M_2 = 010$$

$$M_6 = 110$$

$$M_1 = 001$$

$$M_4 = 100$$

On solving the groupings $F = \bar{C} + \bar{A}B$

Q7). Simplify the Boolean expression $F = A'C + A'B + BC + AB'C$

In sum of product using K-map method.

A \ BC	00	01	10	11
0	000	001	011	010
1	100	101	111	110

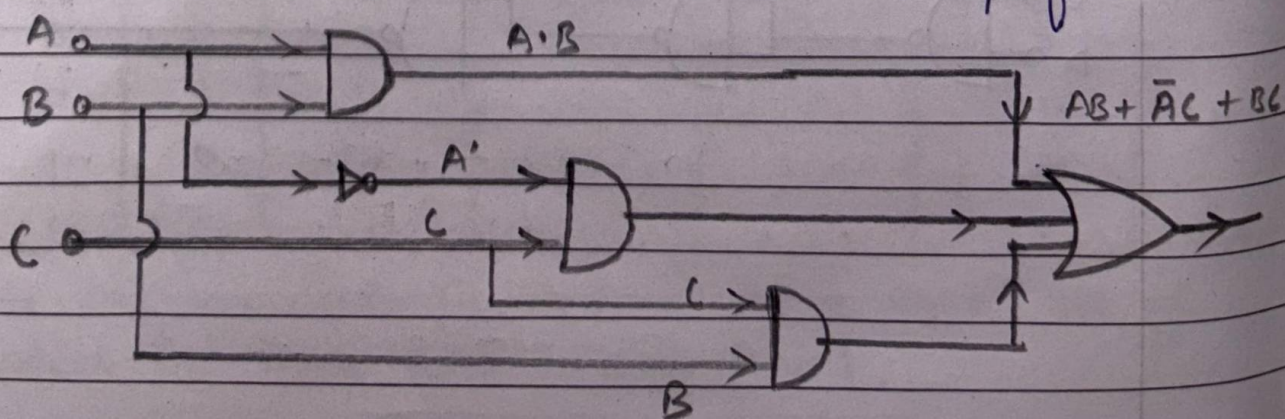
$$F = A'C + A'B + BC + AB'C$$

On simplifying the mid terms $\Rightarrow F = A'C + BC$

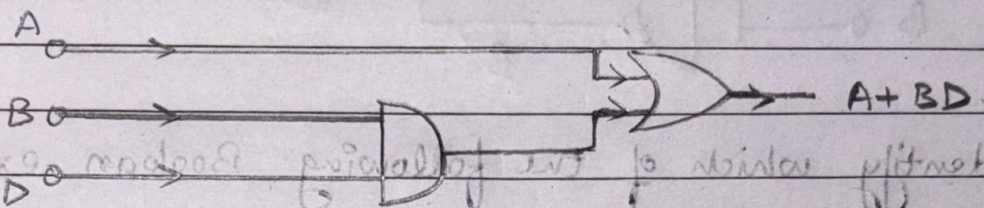
Q8). Using Boolean algebra techniques, simplify the following expressions and realize the minimize expression using basic logic gates.

(a). $AB + \bar{A}C + BC$

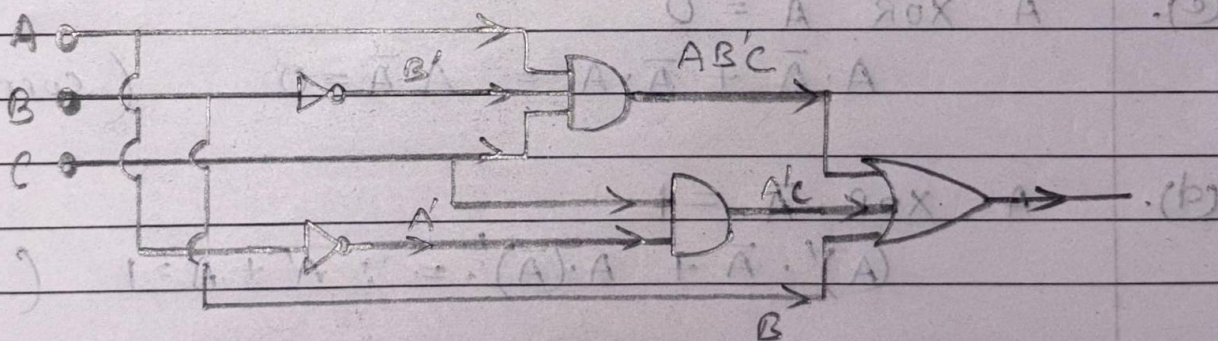
Can't be Simplified.



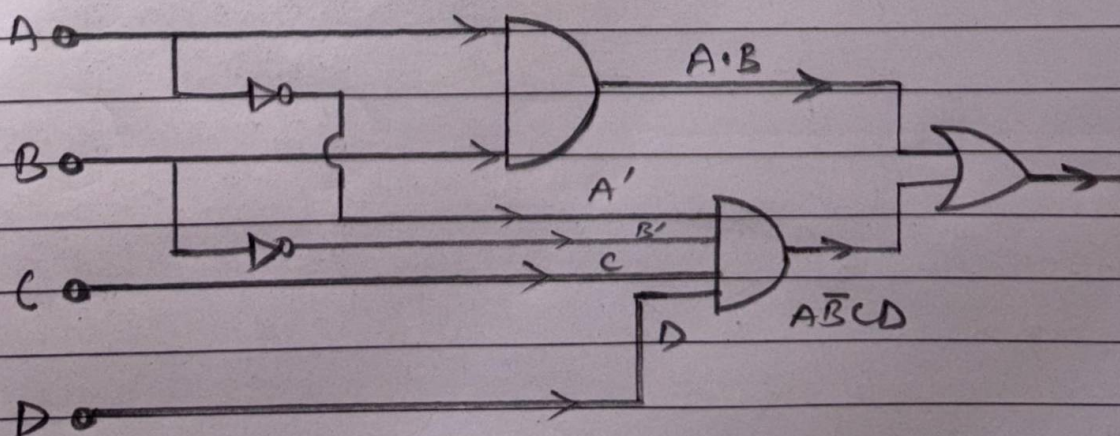
(b). $A + B[AC + (B + \bar{C})D]$
 $= A + B[AC + BD + \bar{C}D]$
 $= A + ABC + BD + \bar{C}BD$
 $= A \cdot (1 + BC) + BD(1 + \bar{C})$
 $= A + BD$



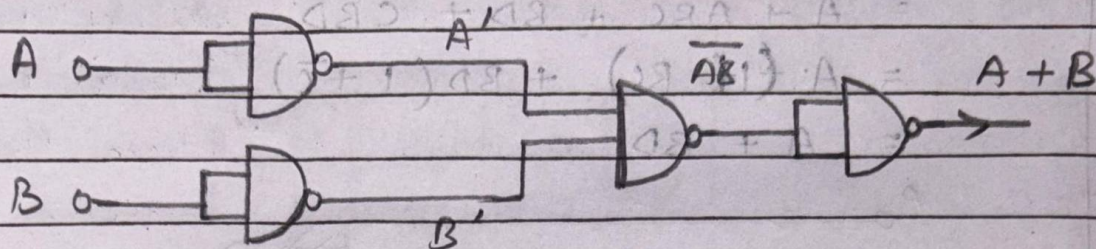
(c). $A\bar{B}C + B + B\bar{D} + AB\bar{D} + \bar{A}C$
 $= A\bar{B}C + B + B\bar{D}(1 + A) + \bar{A}C$
 $= A\bar{B}C + B + B\bar{D} + \bar{A}C = 1$
 $= A\bar{B}C + B(1 + \bar{D}) + \bar{A}C$
 $= A\bar{B}C + B + \bar{A}C$



(d). $ABCD + AB(\bar{C}\bar{D}) + (\bar{A}\bar{B})CD$
 $= AB(CD + \bar{C}\bar{D}) + \bar{A}\bar{B} \cdot CD$
 $= AB(C + \bar{C})(D + \bar{D}) + \bar{A}\bar{B} \cdot CD$
 $= AB + \bar{A}\bar{B} \cdot CD$



Q9): Implement XOR gate using minimum number of NAND gate?



Q10): Identify which of the following Boolean expression is incorrect.

(a). $A \text{ XOR } 0 = A'$

$$\bar{A} \cdot 0 + \bar{0} \cdot A = A + \bar{A} \cdot 0 = A + 0 = A \quad (\text{Incorrect})$$

(b). $A \text{ XOR } 1 = A' \bar{A} + \bar{A} A + A + \bar{A} \bar{A}$

$$\bar{A} \cdot 1 + 1 \cdot A = \bar{A} + A = 1 \quad (\text{correct})$$

(c). $A \text{ XOR } A = 0$

$$A \cdot \bar{A} + \bar{A} \cdot A = A \cdot \bar{A} = 0 \quad (\text{correct})$$

(d). $A \text{ XOR } \bar{A} = 1$

$$(A)' \cdot \bar{A} + A \cdot (\bar{A})' = A' + A = 1 \quad (\text{correct})$$

(e). $A + A'B = A + B \bar{A} \Rightarrow (\bar{A})$ (correct)

(f). $(A+B)(A+C) = A + BC \Rightarrow \bar{A} \bar{B} \bar{C}$ (Distributive Theorem)